Answer Key - Machine Learning - Undergraduate Level Assessment 19 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Linear hard-margin SVM.; B) Linear Logistic Regression.; C) Linear Soft margin SVM.; D) The centroid method.
   Note: Correct option: A - Linear hard-margin SVM.

2. **Answer:** D
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: D - False, True

3. **Answer:** C
   Options: A) 0; B) 1; C) 2; D) 3
   Note: Correct option: C - 2

4. **Answer:** B
   Options: A) they are biased; B) they have high variance; C) they are not consistent estimators; D) None of the above
   Note: Correct option: B - they have high variance

5. **Answer:** A
   Options: A) random crop and horizontal flip; B) random crop and vertical flip; C) posterization; D) dithering
   Note: Correct option: A - random crop and horizontal flip

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Decreases model bias'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'p(y|x, w)'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** def get_perrin(n): | return 3 | return 0 | return 2 | return get_perrin(n - 2) + get_perrin(n - 3)
    Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def Extract(lst): | return [item[0] for item in lst]
    Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key